

RadiologyAI

User Manual

Step-by-Step Guide for Clinical Operators

DA5402 AI Application with MLOps · 2025

Intended Users	Doctors, Radiologists, Hospital Screening Teams
NOT for	Patient self-diagnosis
Application URL	http://localhost:3000
Support	Contact your hospital IT administrator
Version	2.0 · April 2025

⇒ **Important:** This application provides AI-assisted preliminary assessment only. It is NOT a substitute for professional radiological diagnosis. All AI predictions must be reviewed and confirmed by a qualified radiologist before any clinical decision is made.

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Introduction

RadiologyAI is an AI-powered chest X-ray analysis tool designed for clinical operators in emergency rooms, radiology departments, and hospital screening centres. The system analyses a chest X-ray image in under 200 milliseconds and provides:

- A disease classification: **Normal**, **Pneumonia**, or **COVID-19**
- A confidence score showing how certain the AI is
- A risk level: **Low** (Normal) or **High** (Pneumonia / COVID-19)
- A **Grad-CAM heatmap** showing which lung regions the AI focused on
- A **downloadable clinical PDF report** for patient records

⇒ **Important:** RadiologyAI is a decision **support** tool — not a decision **replacement** tool. Always consult a qualified radiologist for final clinical decisions.

Who Should Use This Application

Role	How They Use RadiologyAI
Emergency Room Doctor	Quick triage — flag High Risk cases for immediate radiologist review
Radiologist	Second-opinion tool and explainability review via Grad-CAM
Hospital Screening Team	Batch triage of incoming chest X-rays during high patient volume

Getting Started

System Requirements

Requirement	Specification
Browser	Google Chrome 90+ or Mozilla Firefox 88+ (recommended)
Network	Connected to hospital local network
X-ray format	JPEG (.jpg, .jpeg) or PNG (.png)
Maximum file size	10 MB per image
Screen resolution	1280 × 720 or higher

Opening the Application

Step 1 — Open your browser

Open Google Chrome or Firefox on your workstation.

Step 2 — Navigate to RadiologyAI

Type the following address in the browser address bar and press Enter:

`http://localhost:3000`

If your hospital uses a different server address, contact your IT administrator.

Step 3 — Verify the application loaded

You should see the RadiologyAI home screen with the navigation bar at the top showing five tabs: **Analysis · History · Drift · Models · Monitor**

Tip: If the page shows an error or does not load, contact your IT administrator. The backend server may need to be restarted.

Analysing a Chest X-Ray — Step by Step

This is the main feature of RadiologyAI. Follow these steps to analyse a chest X-ray image.

Step 1 — Upload the X-Ray Image

Method A — Drag and Drop

Open your file manager, locate the chest X-ray image file, and drag it directly onto the dashed upload area in the centre of the screen. The image will appear as a preview when dropped successfully.

Method B — Click to Browse

Click anywhere inside the dashed upload area. A file browser window will open. Navigate to your X-ray image file, select it, and click Open.

Accepted	JPEG (.jpg, .jpeg) and PNG (.png) files up to 10 MB
Not Accepted	DICOM (.dcm), PDF, BMP, TIFF, or files over 10 MB

After a successful upload you will see:

- The X-ray image displayed in the upload area
- The filename shown below the image
- The file size shown in megabytes

⇒ **Important:** If you see the error “*Invalid file type*” — the file format is not supported. Convert the image to JPEG or PNG first. If you see “*File too large*” — the file exceeds 10 MB, compress the image before uploading.

Step 2 — Enter Patient Details (Optional)

Below the upload area you will see two input fields:

- **Patient ID:** Enter the patient’s hospital ID (e.g. PT-000123). This appears in the downloaded PDF report.
- **Age:** Enter the patient’s age in years.

Tip: Patient details are optional but recommended. They appear in the PDF clinical report and help identify the scan in the History page.

Step 3 — Run the Analysis

Click the Analyse X-Ray button

Click the dark blue “Analyse X-Ray” button below the patient details fields.

During analysis you will see:

- The button text changes to “Analysing...”
- A loading indicator appears
- On the GPU server this takes under 1 second
- On a CPU-only machine this may take up to 30 seconds

⇒ **Important:** Do not close the browser tab or navigate away while the analysis is running. Wait for the results to appear.

Understanding Your Results

After analysis completes the Results page appears automatically with four sections.

Section 1 — Diagnosis Card

The coloured card on the right shows the AI diagnosis:

Result	Card Colour	Risk Level	Clinical Meaning
Normal	Green	Low	No significant pulmonary pathology detected
Pneumonia	Red	High	Bacterial or viral lung infection detected
COVID-19	Red	High	COVID-19 related lung changes detected

The card also shows:

- **Confidence:** How certain the AI is (e.g. 89.3% = high confidence)
- **Confidence bar:** Visual representation of certainty
- **Inference Time:** How long the analysis took

⇒ **Important:** A **Normal** result does not guarantee the patient is disease-free. It means the AI did not detect patterns associated with Pneumonia or COVID-19. Always apply clinical judgement.

Section 2 — Class Probabilities Chart

The horizontal bar chart shows the AI's confidence for all three possible diagnoses simultaneously. All three bars together add up to 100%.

- The longest bar = the predicted class
- A confident prediction has one very long bar and two very short bars
- If two bars are similar in length, the AI is uncertain — apply extra clinical caution

Section 3 — Grad-CAM Heatmap

The heatmap image shows which regions of the lung the AI focused on:

Colour	Meaning
Red / Orange	Highest AI attention — these regions most influenced the prediction
Yellow / Green	Moderate AI attention
Blue / Dark	Low AI attention — AI did not consider these regions important

Tip: Use the Grad-CAM to verify the AI is focusing on clinically relevant areas. If the AI predicts Pneumonia but focuses on non-lung areas (e.g. the spine or rib borders), apply additional clinical scrutiny.

Section 4 — X-Ray Preview

The original uploaded X-ray image is shown on the left for direct comparison with the Grad-CAM heatmap.

Downloading the Clinical PDF Report

After reviewing the AI results, you can download a formatted clinical PDF report for patient records.

Step 1 — Click the Download PDF Report button

Scroll down on the Results page. Click the dark “Download PDF Report” button.

Step 2 — Wait for PDF generation

The PDF is generated automatically in 2–5 seconds. A download dialog will appear in your browser.

Step 3 — Save the PDF

Save the file to your preferred location. The filename format is:
`RadiologyAI_Report_{PatientID}_{Timestamp}.pdf`

What the PDF Contains

Section	Content
Header	Patient ID, date/time, report ID, model used
Diagnosis	Predicted class, confidence, risk level in large formatted text
Class Probabilities	Table showing confidence for all 3 classes
Grad-CAM Heatmap	The explainability heatmap image embedded in the PDF
Inference Details	Model name, inference time, image size, framework
Disclaimer	Mandatory clinical disclaimer text

Tip: Print the PDF and attach it to the patient’s physical file, or upload it to the hospital’s electronic health record system.

Providing Radiologist Feedback

After reviewing the AI prediction clinically, please provide feedback. Your feedback improves the AI model over time.

Step 1 — Review the AI prediction clinically

Apply your clinical expertise to assess whether the AI prediction is correct.

Step 2 — Add optional comments

If you wish, type the correct diagnosis or clinical notes in the text box above the feedback buttons.

Step 3 — Click the appropriate button

- Click the **green “Confirm Prediction” button** if the AI diagnosis is correct
- Click the **red “Incorrect” button** if the AI diagnosis is wrong

Step 4 — Confirmation

A green notification appears: *“Feedback submitted — thank you!”*

The message also shows the updated model accuracy across all feedback submissions.

Why feedback matters:

Your feedback directly drives model improvement. The system monitors accuracy from all radiologist feedback in real time. If accuracy drops below 80%, an automatic alert is sent and model retraining is triggered. Your clinical expertise makes the AI better over time.

Starting a New Scan

Click the New Scan button

Click the “New Scan” button at the top right of the Results page. The upload area resets and you can analyse a new X-ray immediately.

Tip: You can also click “Analysis” in the navigation bar at the top to return to the upload page at any time.

Patient History Page

The History page shows all past X-ray scans and radiologist feedback submissions.

Viewing Scan History

1. Click **History** in the top navigation bar
2. A table appears showing all past scans: timestamp, prediction ID, diagnosis, confidence, and radiologist confirmation status
3. The most recent scans appear at the top

Filtering by Diagnosis

Click the filter buttons above the table to show only scans of a specific class:

- **All** — shows every scan
- **Normal** — shows only Normal predictions
- **Pneumonia** — shows only Pneumonia predictions
- **COVID19** — shows only COVID-19 predictions

Viewing Scan Detail

Click any row in the table to expand a detail panel on the right side showing:

- Full diagnosis card with confidence
- Exact timestamp
- Full prediction ID
- Radiologist confirmation status
- Correct class (if marked incorrect)
- Radiologist comments

Summary Statistics

At the bottom of the History page, four cards show summary counts:

- Total Scans performed
- Number of Normal predictions
- Number of Pneumonia predictions
- Number of COVID-19 predictions

Drift Detection Page (Admin)

The Drift Detection page shows whether the X-ray data distribution has changed compared to the training baseline. This is relevant for hospital administrators and radiologists monitoring AI reliability.

1. Click **Drift** in the top navigation bar
2. Four cards appear: Overall Status + one card per class (Normal, Pneumonia, COVID19)
3. Each class card shows a drift score (0.0 to 1.0) and status:
 - **Green / Stable** — data distribution is similar to training data
 - **Red / Drift Detected** — significant distribution shift detected
4. Click any class card to load the full Evidently AI drift report in the panel below
5. The report shows detailed distribution plots for each pixel statistic

What does drift mean?

If the hospital’s X-ray machine settings change, or X-rays from a different machine are uploaded, the pixel distribution may differ from the training data. High drift may cause the AI accuracy to degrade. Contact your IT administrator to trigger model retraining.

Model Comparison Page (Admin)

The Model Comparison page shows all AI training experiments and allows comparison across different model versions.

- 1. Click **Models** in the top navigation bar
- 2. The best model is highlighted in green at the top with its key metrics
- 3. The bar chart shows all runs compared on the selected metric
- 4. Use the dropdown to switch between metrics: val_f1, val_acc, test_f1, macro_auc
- 5. The table below shows all runs with full hyperparameter and metric details

Monitor Page (Admin)

The Monitor page shows the system health and ML infrastructure status.

Panel	What It Shows
API Status	Online/Error — whether the backend is responding
Model Loaded	Yes/No — whether EfficientNetB0 is loaded in memory
Model	EfficientNetB0 (model architecture)
Framework	PyTorch 2.3.0
Classes	3 (Normal, Pneumonia, COVID19)
Image Size	224 × 224 pixels
Grafana Dashboard	Click link to open live metrics dashboard
MLflow Experiments	Click link to view all training runs
DVC Pipeline	Shows the 9-stage data pipeline diagram

Troubleshooting

Problem	Solution
Page does not load	Check your network connection. Ensure you are connected to
“Model not loaded” error	The backend is starting up. Wait 30 seconds and click Refresh
Analysis takes very long	On the GPU server, analysis takes under 1 second. On a CPU-only machi
“Invalid file type” error	Only JPEG (.jpg, .jpeg) and PNG (.png) files are accepted. If you
“File too large” error	The file exceeds 10 MB. Use image editing softw
Grad-CAM heatmap not showing	This occasionally occurs for unusual image formats or very small in

Problem	Solution
PDF download fails	Try refreshing the page and repeating the analysis
Drift page shows “not run”	The drift detection has not been run yet.
No history showing	The History page shows only scans for which radiologist feedback was submitted
Feedback not submitting	Ensure you are connected to the network. Check your internet connection

Quick Reference Card

Task	How To Do It
Analyse X-ray	Analysis tab → Upload image → Click Analyse X-Ray
Download PDF report	Results page → Click Download PDF Report button
Give feedback	Results page → Scroll down → Click Confirm or Incorrect
View past scans	Click History tab in navbar
Filter history by class	History page → Click Normal / Pneumonia / COVID19 buttons
View drift report	Click Drift tab → Click class card → View Evidently report below
Compare models	Click Models tab → Use metric dropdown → View bar chart
Check system health	Click Monitor tab → Check API Status panel
Start new scan	Results page → Click New Scan button

Clinical Disclaimer

Clinical Disclaimer

RadiologyAI provides AI-assisted preliminary assessment only. It is NOT a substitute for professional medical diagnosis. All AI predictions must be reviewed and confirmed by a qualified radiologist before any clinical decision is made. This system is intended as a triage decision support tool to assist — not replace — clinical expertise.

Patient X-ray images are processed for inference only and are not stored permanently by this system. Feedback data is stored locally for model improvement and is never shared with third parties.

In case of medical emergency, do not rely on AI results alone. Follow established emergency clinical protocols.